

Overview of Future Skills and Cyber Security



It is an industry utility to get India accelerated on a journey to build to its skills and become the global hub for talent in the emerging technologies.

There are 9 Skills covered for this initiative and they are:

- Internet of Things.
- Big Data Analytics.
- Cloud Computing .
- Virtual Reality.
- Artificial Intelligence.
- Social & Mobile .
- Block chain Technology.
- 3D printing/Additive Manufacturing .
- Robotic Process Automation.

- It refers to a new industrial technology era that is transforming current systems, sensors, machines and workloads. The 'fourth industrial revolution' was first introduced by **Klaus Schwab** in 2015 in the article in Foreign Affairs.
- First industrial Revolution-1765
- The Second industrial Revolution-1870
- The Third Industrial Revolution-1969
- Fourth industrial Revolution –Now a Days.

- It is the extension of Internet connectivity into physical devices and everyday objects.
- Embedded with electronics, Internet Connectivity and other forms of hardware, these devices can communicate and interact with others over the internet , and they can be remotely monitored and controlled.

➤ **Applications of IoT**

Smart Homes. One of the best and the most practical applications of IoT, smart homes really take both, convenience and home security, to the next level. ...

- Smart City. ...
- Self-driven Cars. ...
- IoT Retail Shops. ...
- Farming. ...
- Wearables. ...
- Smart Grids. ...
- Industrial Internet.

The main goal of the big data analytic is to help organization to make better business decision, future prediction, analysis large numbers of transactions that done in organization and update the form of data that organization is used.

Ex-Big online business website like Flipkart, snapdeal uses facebook or Gmail data to view the customer information or behavior.

- **Velocity**-The rapid generation of data from time to time is called velocity.
- **Volume**-it means storage capacity.
- **Variety**: the diversity and range of different data types, including unstructured data, semi-structured data and raw data.
- **Veracity**: the “truth” or accuracy of data and information assets, which often determines executive-level confidence.
- **Value**: the most important “V” from the perspective of the business, the value of big data usually comes from insight discovery and pattern recognition that lead to more effective operations, stronger customer relationships and other clear and quantifiable business benefits

- **Travel and Tourism**
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- Healthcare
- Education
- Government
- Media
- Insurance

- Introduction to cloud-computing means storing and accessing data and programs over the internet instead of your computer's hard drive which comes to an existence in 1950's.
It provides services both public and private networks. Ex-E-mail, GoogleDrive etc.

Types of Cloud

- Private Cloud-it delivered from a business data center to internet users.
- Public and Community Cloud-it is third party cloud service provider delivers the cloud service over the internet.
- Hybrid Cloud-it is combination of Public and Private Cloud.

- Infrastructure as a Service(IaaS)-it is the most basic level of service which provides virtualized computing resources over the net.
- Platform as a Service(PaaS)- it provides host development tools on their infrastructures.
- Software as a Service(SaaS)-it is distribution model which delivers s/w applications over the internet.

- Online Data Storage
- Backup and Recovery
- Testing and Development
- Big Data Analysis
- Antivirus applications
- In E-Commerce.

- (VR) is a computer-generated environment with scenes and objects that appear to be real, making the user feel they are immersed in their surroundings. This environment is perceived through a device known as a Virtual Reality headset or helmet.



- Non-Immersive-This technology provides a computer-generated environment, but allows the user to stay aware of and keep control of their physical environment.
- Semi Immersive-Semi-immersive virtual experiences provide users with a partially virtual environment. This category of VR is used often for educational or training purposes A video game is a great example of a non-immersive VR experience.

- Fully-immersive simulations give users the most realistic simulation experience, complete with sight and sound. To experience and interact with fully-immersive virtual reality, the user needs the proper VR glasses or a head mount display (HMD). This type of VR has been commonly adapted for gaming and other entertainment purposes, but usage in other sectors,

- Augmented reality is an interactive experience that combines the real world and computer-generated content.
- AR is used in Medical, Education, Entertainment etc.

- It is a branch of computer science named AI pursues creating the computers or machines as intelligence as human beings. 1956 - John McCarthy coined the term 'artificial intelligence' and had the first AI conference.

Types of Artificial Intelligence

Purely Reactive

- These machines do not have any memory or data to work with, specializing in just one field of work. For example, in a chess game, the machine observes the moves and makes the best possible decision to win.

Limited Memory

- These machines collect previous data and continue adding it to their memory. They have enough memory or experience to make proper decisions, but memory is minimal. For example, this machine can suggest a restaurant based on the location data that has been gathered.

Theory of Mind

- This kind of AI can understand thoughts and emotions, as well as interact socially. However, a machine based on this type is yet to be built.
- Self-Aware
- Self-aware machines are the future generation of these new technologies. They will be intelligent, sentient, and conscious.

- Healthcare
- Handwriting Recognition
- Gaming
- Manufacturing Sector
- Intelligent Robots
- Speech Recognition
- Language Translations
- Intelligent Robots

- Block chain technology is an advanced database mechanism that allows transparent information sharing within a business network. A blockchain database stores data in blocks that are linked together in a chain.
- Application of Block Chain
 - Money Transfer
 - Financial Exchanges
 - Insurance
 - In Real State
 - Voting etc.

- Robotic process automation is a form of business process automation technology based on metaphorical software robots or on artificial intelligence /digital workers. It is sometimes referred to as software robotics.
- Applications of RPA
 - Conduct High Volume repeated tasks
 - Perform multiple tasks
 - Automated report generation
 - Technical debt management
 - Product management etc.

- Cyber security refers to every aspect of protecting an organization and its employees and assets against cyber threats.
- Types of cyber security
- Attacks on confidentiality-these includes stealing your personal identifying information and your bank account or credit card information.
- Attacks on integrity- in this a cybercriminal will access and release sensitive information for the purpose of exposing the data and influencing the public to lose trust in that organization.
- Attacks on availability- the aim of this type of cyber attack is to block users from accessing their own data until they pay a fee or ransom.

- Only use trusted sites when providing your personal information.
- Don't open e-mail attachments or click links in e-mails from unknown sources.
- Always keep your devices updated.
- Enable your firewall.
- Install a good Antivirus.
- Give your PC a strong password.

- only buy smart phones from trusted vendors.
- Do not save all passwords.
- Make sure your Wi-fi is Secured.
- Use the Android Security App.
- Backup your Android phone's data .
- Install trusted Apps from play store.
- Use always an original battery charger.